

Reinforcement Learning from Ecological Feedback (RLEF)

A Framework for Aligning Language-Model Planning with Planetary Boundaries

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Outline

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Motivation

Plans that read well — but act on a planet

Language models can generate and reason over plans [9], so they are increasingly considered for **energy**, **infrastructure** and **climate policy** — though their planning abilities remain contested [27].

The core difficulty: a plan's consequences are **delayed and systemic**.

- ⚠️ They unfold over long horizons, through coupled physical, economic and social systems [22].
- ⚠️ Choices made this decade shape climate risks for generations [11].
- ⚠️ A carbon-market rule can *read* as reasonable, yet misfire once its effects propagate.

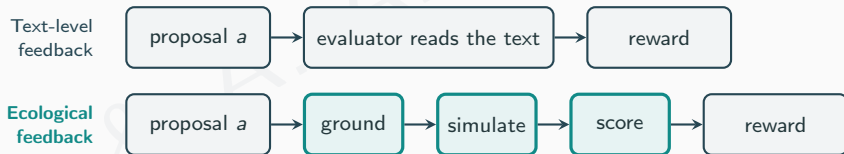
The question

Can the modeled **physical consequences** of a plan become part of the **alignment signal** itself?

Today's alignment judges the text

Most alignment learns from judgments over candidate outputs — **RLHF** [16], **DPO** [18], **Constitutional AI** [2].

- ✓ They can represent environmental values.
- ✗ But the evaluator must *infer* consequences that may depend on decades of system dynamics.



💡 A pattern related to execution feedback for code [7]: **execute, then score** — here with an environmental simulator as the executor.

Text-level vs. modeled feedback: three implications

1 Long horizons

Consequences unfold over **decades**; valuing them is itself contested [15, 5]. Text-level evaluation leaves the forecast to the evaluator.

2 Bounded evaluation

Biodiversity loss, freshwater stress, cumulative emissions are properties of **trajectories** — not of a proposal's wording.

3 Misspecification

Explicit modeling does not end *reward hacking* [24]. **RLEF relocates** that risk — into assumptions that are more explicit and inspectable.

The separation RLEF makes

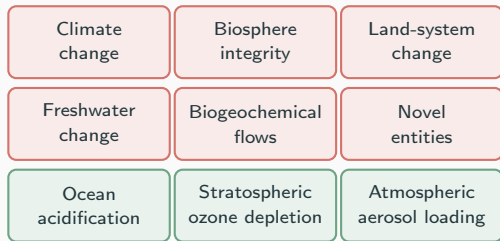
Which outcomes to value stays a human and societal judgment. **What a plan will physically do** is where external models can add evidence.

1. **Diagnosis** — why text-level preference feedback falls short when consequences are delayed and mediated by physical systems.
2. **RLEF** — a simulator-mediated feedback architecture; its penalty objective corresponds to a **fixed-multiplier Lagrangian relaxation** of constrained RL.
3. **Illustration** — how **RLEF** could be instantiated for international carbon-market design.
4. **Open problems** — grounding, simulator fidelity and uncertainty, computational cost, normative specification, empirical validation.

 A conceptual and formal framework — not an empirically validated training method.

Planetary Boundaries as Constraints

Nine boundaries — six already crossed



□ transgressed □ not transgressed (2023)

- The **Planetary Boundaries** (PB) framework [21, 25] sets control variables and boundary levels for a **safe operating space**.
- Crossing a boundary is not instant collapse — it enters a **zone of increasing risk**.
- 2023 assessment: **6** of **9** transgressed globally [19].

Implication for feedback

Planning starts *outside* the safe space, so feedback must reward **movement back** — not only avoiding new crossings.

One number is not enough

Mitigation pathways that lean heavily on **biomass-based negative emissions** [10] can have these effects:

Boundary	Effect
Climate change	✓ atmospheric carbon reduced
Land-system change	✗ pushed beyond its boundary
Biosphere integrity	✗ pushed beyond its boundary
Biogeochemical flows	✗ pushed beyond its boundary

A carbon-only objective...

...sees only the first row.

Planetary boundaries let a planner track **several ecological variables** against their boundary levels at once.

⚠ But PB is *not* a complete objective: welfare, feasibility and burden sharing must be set explicitly.

Safe and just: keep physics and values apart

Safe and just Earth-system boundaries [20] add a justice dimension: an aggregate outcome can meet a global safety criterion while exposing particular populations to unacceptable harm.

Biophysical constraints

→ Enter as **boundary terms**: control variables — the ecological state along each trajectory.

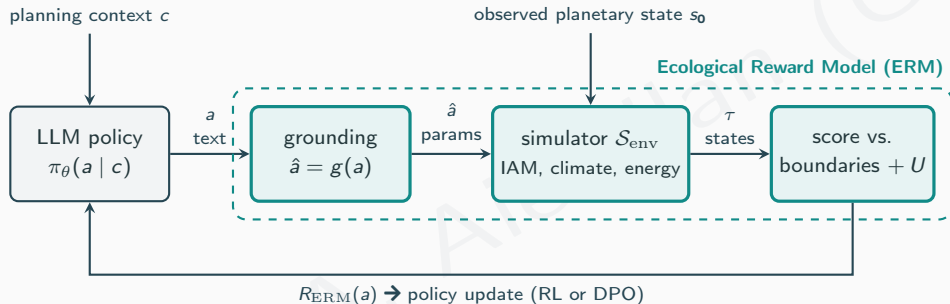
Distributive objectives

→ Enter the **utility term**: energy access, regional welfare, development needs, transition costs.

- For climate and atmospheric aerosol loading, the justice criterion gives a **more stringent** boundary than the safety criterion.
-  PB itself is revised over time: tie each constraint to specific models, versions and uncertainty estimates.

The RLEF Framework

RLEF at a glance



- The **ERM** is *not* fitted to preference data: it is a computational pipeline — ground, simulate with a domain model (e.g. an integrated assessment model, IAM), score.
- 💡 The contribution is the **feedback architecture**, not a new policy-optimization algorithm.

Problem formulation

- Context $c \in \mathcal{C}$: objective, observations, stated constraints.
- Initial state s_0 : set from **observed** control variables.
- Policy $\pi_\theta(a \mid c)$ proposes $a \in \mathcal{A}$: a **complete policy** — e.g. a carbon price, subsidies, regulations and a schedule — not a per-step action.
- Grounding $g : \mathcal{A} \rightarrow \hat{\mathcal{A}}$ gives a simulator-executable intervention $\hat{a} = g(a)$.
- Simulator: $\tau(\hat{a}) = (s_0, s_1, \dots, s_H)$. (Deterministic for notation; a stochastic one adds an expectation.)

Grounding is part of alignment

RLEF scores the consequences of $\hat{a}$, not the wording of a .

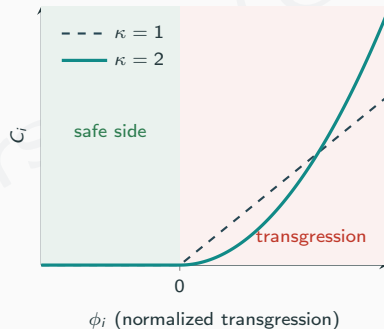
If $g(a)$ loses the proposal's meaning, the feedback can mislead —
even with an accurate simulator.

Scoring a trajectory: the violation cost

Violation cost for boundary $i \in \mathcal{I}$

$$C_i(s_t) = \left[\max(0, \phi_i(B_i(s_t); L_i)) \right]^\kappa$$

- $\mathcal{I} \subseteq \{1, \dots, 9\}$: boundaries the simulator represents; the rest give *no* feedback — declare them.
- B_i control variable, L_i boundary specification.
- $\phi_i \leq 0$ on the safe side, > 0 and growing beyond it — its normalization is a **modeling choice**.
- $\kappa \geq 1$; with $\kappa > 1$ the marginal penalty rises with transgression.



Shape of the violation cost (illustrative κ values).

The ecological reward

Reward of proposal a , along $\tau(g(a))$

$$R_{\text{ERM}}(a) = \sum_{t=0}^H \gamma^t \left[\underbrace{U(s_t, \hat{a})}_{\text{task utility}} - \underbrace{\sum_{i \in \mathcal{I}} w_i C_i(s_t)}_{\text{ecological penalty}} \right]$$

$\gamma \in (0, 1]$: temporal weight | $w_i \geq 0$: penalty weights | U : e.g. economic output, energy access

→ **Ecological state** enters through the boundary variables $B_i(s_t)$.

→ **Value judgments** sit in U , γ , w_i , κ , ϕ_i — where they can be inspected and contested.

Starting already transgressed

For a transgressed boundary $C_i(s_0) > 0$ — but it is the same for every candidate from that s_0 , so it does not change the ranking. What differs: does a trajectory **reduce** or **deepen** the transgression?

Plugging the ERM into policy optimization

Route 1 — KL-regularized RL

As in **RLHF** [16]:

$$J(\theta) = \mathbb{E}_{c \sim \mathcal{D}} \left[\mathbb{E}_{a \sim \pi_{\theta}(\cdot | c)} [R_{\text{ERM}}(a)] - \beta D_{\text{KL}}(\pi_{\theta}(\cdot | c) \| \pi_{\text{ref}}(\cdot | c)) \right]$$

$\beta > 0$ limits departures from the instruction-tuned π_{ref} .

Route 2 — preference optimization

Sample several plans → ground → simulate → score; pair higher- against lower-scoring plans for **DPO** [18].

The ordering comes from **modeled consequences**, not from an evaluator's judgment.

💡 The **ERM** scores *trajectories*, so **observed** outcomes of implemented interventions can be scored the same way — a check on the simulator.

Iterative RLEF with consequence-derived preferences

Algorithm 1 (sketch)

$\pi_{\theta_0} \leftarrow \pi_{\text{ref}}$; **for** rounds $k = 0, \dots, K - 1$: $\mathcal{B}_k \leftarrow \emptyset$

for each context $c \in \mathcal{D}$:

1. sample $a_1, a_2 \sim \pi_{\theta_k}(\cdot \mid c)$
2. ground $\hat{a}_j \leftarrow g(a_j)$
3. simulate $\tau_j \leftarrow \mathcal{S}_{\text{env}}(s_0, \hat{a}_j)$
4. score $R_{\text{ERM}}(a_j)$ from τ_j
5. **if** scores differ: add (c, a^+, a^-) to $\mathcal{B}_k$

DPO update on $\mathcal{B}_k$, reference $\pi_{\theta_k} \rightarrow \pi_{\theta_{k+1}}$

- $\rightarrow$ Feedback is **regenerated every round**: policy updates shift the distribution of plans.
- $\rightarrow$ The current policy is the DPO reference.
- $\rightarrow$ Ties are discarded — they carry no preference information.
- $\rightarrow a^+ / a^-$: higher/lower-scoring plan; $\mathcal{B}_k$: the round's preference buffer.

The penalty as a fixed-multiplier Lagrangian relaxation

Expected discounted utility and violation cost, over $c \sim \mathcal{D}$, $a \sim \pi_\theta(\cdot \mid c)$:

$$J_U(\theta) = \mathbb{E} \left[\sum_{t=0}^H \gamma^t U(s_t, \hat{a}) \right], \quad J_{C_i}(\theta) = \mathbb{E} \left[\sum_{t=0}^H \gamma^t C_i(s_t) \right]$$

Constrained problem

$$\max_{\theta} J_U(\theta) \quad \text{s.t.} \quad J_{C_i}(\theta) \leq d_i, \quad i \in \mathcal{I}$$

$d_i \geq 0$: tolerated discounted violation cost.

Lagrangian

$$\mathcal{L}(\theta, \mu) = J_U(\theta) - \sum_{i \in \mathcal{I}} \mu_i (J_{C_i}(\theta) - d_i), \quad \mu_i \geq 0$$

By linearity of expectation:

$$\mathbb{E}[R_{\text{ERM}}(a)] = J_U(\theta) - \sum_{i \in \mathcal{I}} w_i J_{C_i}(\theta)$$

So, with $\mu_i = w_i$ held fixed

Maximizing $\mathcal{L}$ over θ is maximizing the expected ecological reward — up to the constant $\sum_i w_i d_i$, which does not depend on θ .

What the connection does — and does not — say

	Penalty form (RLEF)	Constrained form
You specify	weights w_i	bounds d_i on violation cost
Utility vs. violations	traded at rates fixed in advance	bounds enforced by adapting μ_i during optimization
Optimizer	KL-regularized RL or DPO	Lagrangian methods, or CPO [1]

-  Related, *not* equivalent formulations.
- For a boundary already transgressed at s_0 , $d_i = 0$ is infeasible: d_i is a **restoration target**, not a prohibition.
- At the policy level this is a **constrained contextual bandit**: one complete proposal per context; the horizon H sits inside each evaluation.

Illustrative Example: Carbon Markets

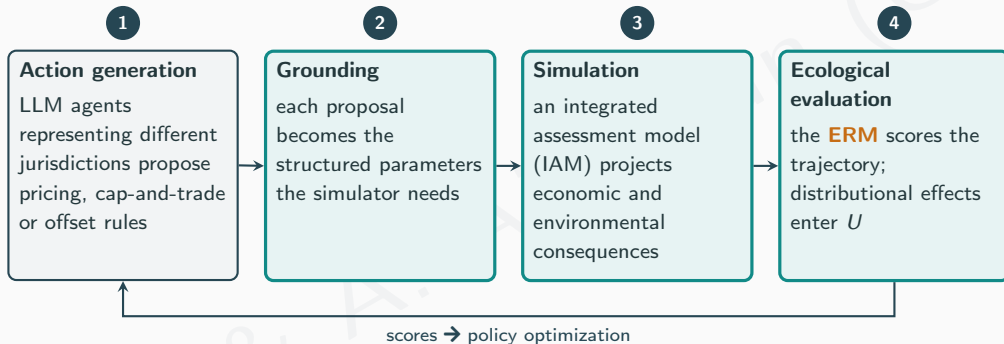
- **Article 6.2:** internationally transferred mitigation outcomes used toward nationally determined contributions [26].
- **Article 6.4:** a mechanism for mitigation and sustainable development [26].
- Rules can be judged by their **emissions trajectories** and their **distributional consequences**.

Why a language model here?

Proposals are **heterogeneous and compositional** — pricing rules, exemptions, schedules, distributional provisions — and must be grounded before they can be simulated.

 If the interventions are already a compact set of numbers, direct numerical optimization may be preferable.

A conceptual RLEF pipeline in four stages



⚠ Several negotiating agents make this a **multi-agent extension**; its interaction dynamics are not formalized here.

The simulator is part of the specification

	DICE [14]	GCAM [3]
Resolution	globally aggregated	regions and sectors
Represents	economy, emissions, carbon cycle, climate	energy, water, land, climate, economy
Blind spot	regional and sectoral structure of many international mechanisms	not necessarily every boundary control variable; missing ones fall outside $\mathcal{I}$

- ⚠ IAM assumptions on technology, damages, behavior and discounting are uncertain and, in places, contested [17, 15].
- ⚠ Responses come from optimization or market equilibrium, not the strategic interaction of learning agents in the **AI Economist** [28] — so gaming by participants stays out of view.

Normative Tensions

Intergenerational fairness vs. near-term costs

U , γ and w_i jointly set the trade-off between near-term welfare and long-run consequences — with distinct roles:

- boundary constraints decide **which** outcomes are penalized;
- γ weights every outcome by **when** it occurs.

Boundaries alone do not fix a social discount rate.

- 💡 **RLEF** does not settle this — it makes the choices **explicit** and contestable.
- Just transitions: concentrated costs on carbon-intensive communities must count [13] — in U or in added constraints.

Shared concern, disputed means

Longtermism: great moral weight on future populations [8].

Growth: sustained growth is argued to be the most reliable way to benefit them [4].

The Global North–South divide

- ✗ A planner optimized only for aggregate emissions reduction could favor restricting industrial expansion in lower-income regions — deeply inequitable.
- **Subsistence** vs. **luxury** emissions [23]: not every unit of emissions is morally interchangeable.
- So the **ERM** must also represent distributional consequences: historical responsibility, energy access, regional welfare, transition burdens — via U or separate constraints.

Not a technical question

Societies and stakeholders may reasonably disagree on which criteria to adopt.

Formalization shows *what* is optimized — it does not replace political deliberation.

Discussion & Conclusion

Where the risk lands

RLEF moves specification risk from text-level judgments into models and interfaces:

Grounding

The **ERM** scores $\hat{a} = g(a)$, not a . An optimized policy could exploit weak translation; the KL term does not check that a and $\hat{a}$ agree.

Simulator misspecification

Optimizing repeatedly against one simulator can favor policies that exploit its structural assumptions.

No empirical validation

The example is illustrative. **RLEF** is a **research framework**, not a deployment-ready method.

Open challenges and future directions

Computational cost

Simulators are expensive: cheap reduced-order models or learned surrogates early, high fidelity for promising or uncertain plans. Surrogates sped up weather forecasting [12] — climate-economy fidelity is open.

Grounding

Constrained formats, typed action schemas, semantic consistency checks — tested against what the simulator actually executes.

Model uncertainty

Score across ensembles of models or parameterizations, with worst-case or risk-sensitive objectives [6].

Evaluation protocol

Scenarios, executable action spaces, simulators, ecological metrics and robustness tests — versus preference-based, constrained-RL and non-learning baselines.

1. ★ **Consequences into the signal** — ground, simulate, and score plans against planetary boundaries plus a task utility.
2. ★ **Physics and values kept apart** — ecological state via boundary variables; value judgments in $U, \gamma, w_i, \kappa, \phi_i$.
3. ★ **Architecture, not algorithm** — the penalty objective can be viewed as a fixed-multiplier Lagrangian relaxation of constrained RL.

The open empirical question

Does simulator-mediated feedback improve planning behavior — without simply shifting reward misspecification into the grounding and simulation pipeline?

Thank you!

Questions?

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Backup: the observational channel

Nothing in the **ERM** requires a *simulated* trajectory. The observed ecological trajectory after an implemented intervention can be scored the same way.

- ✓ Realized outcomes test the simulator's predictions and can inform its calibration.
- ✗ Observation exists only for implemented interventions, and only after the relevant delays.

Division of labor

Simulation supplies feedback on counterfactual proposals that training needs.

Observation supplies evidence about whether the simulator is right.

Backup: notation at a glance

$c, \mathcal{D}$	context; context distribution	$\mathcal{I}$	boundaries the simulator represents
s_0	initial state (observed)	B_i, L_i	control variable; boundary spec.
$\pi_\theta, \pi_{\text{ref}}$	LLM policy; reference policy	ϕ_i, κ	normalized transgression; exponent
$a, \hat{a} = g(a)$	proposal; grounded intervention	C_i, w_i	violation cost; penalty weight
$\mathcal{S}_{\text{env}}, \tau, H$	simulator; trajectory; horizon	U, γ	stage utility; temporal weight
β	KL-regularization strength	d_i, μ_i	tolerated cost; Lagrange multiplier
$a^+, a^-, \mathcal{B}_k$	preferred / rejected plan; buffer	J_U, J_C	expected discounted utility; cost

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